

ALOK KUMAR RATHAUR

Senior iOS Engineer | Swift, SwiftUI & Apple Platform Architecture

Kanpur, India | Open to Relocation / Hybrid / Remote

alok.rathaur26@gmail.com | +91-7571868833

LinkedIn: [linkedin.com/in/alokrathaur](https://www.linkedin.com/in/alokrathaur) | GitHub: github.com/alok-rathaur | Portfolio: alokrathaur.github.io

PROFESSIONAL SUMMARY

Senior iOS Engineer with 8.3+ years of software engineering experience (7+ years native Apple ecosystem) architecting, developing, and deploying high-impact iOS and macOS applications across Fintech, E-Commerce, and Telehealth. Deep expertise in Swift 5/6, UIKit, SwiftUI, Swift Concurrency (async/await, Actors), Core Data, and Combine. Proven track record implementing scalable mobile architectures (MVVM-C, Clean Architecture, VIPER, TCA), delivering production outcomes including 35% crash rate reduction, 30% faster cold launch times, and 40% lower memory footprint. Extensive domain experience in high-security European multi-currency payment systems integrating Apple Pay, Payment SDKs, and 3D Secure flows, as well as high-volume retail e-commerce (100K+ downloads). Demonstrates complete technical ownership from product requirements and API design to CI/CD pipelines (GitHub Actions, Fastlane), App Store distribution, and production monitoring.

CORE TECHNICAL SKILLS

iOS & Apple Platforms: Swift (5/6), Objective-C, UIKit, SwiftUI, AppKit, Cocoa Touch, Foundation, Combine, Core Data, Apple Pay, StoreKit 2 (In-App Purchases / Subscriptions), XCTest, XCUITest, Xcode Instruments

Payment & Financial Systems: Apple Pay, Stripe, Checkout.com, Worldpay, StoreKit 2, Dodo Payments, 3D Secure (3DS), PSD2 Strong Customer Authentication (SCA), PCI-DSS Guidelines, Multi-Currency FX Checkout

Swift Concurrency & Threading: Swift Concurrency, async/await, Actors, Structured Concurrency, Tasks & Task Groups, Grand Central Dispatch (GCD), Operations, ARC & Retain Cycle Elimination

Architecture & Patterns: MVVM, MVVM-C (Coordinator Pattern), Clean Architecture, VIPER, MVC, The Composable Architecture (TCA), ReactorKit, SOLID Principles, Protocol-Oriented Programming (POP), Dependency Injection

Performance & Debugging: Xcode Instruments (Time Profiler, Allocations, Leaks), Memory Graph Debugger, LLDB, Launch Time Optimization, Binary Size Reduction, Crashlytics & Root-Cause Analysis

Networking & Security: RESTful APIs, JSON Parsing, URLSession, Alamofire, WebSockets, API Contract Design, Local Caching & Offline Sync, Secure Enclave, Keychain, OWASP Mobile Security

CI/CD & Release: Xcode, Fastlane, GitHub Actions, Bitrise, Git, GitHub, TestFlight, App Store Connect, Code Signing & Provisioning Profiles, Swift Package Manager (SPM), CocoaPods

Analytics & SDKs: Firebase (Analytics, Crashlytics), UXCam, Coralogix, Branch SDK (Deep Linking), WebEngage (Push Notifications), Lottie Animations

Cross-Platform & Full-Stack: React Native (Expo), TypeScript, Next.js, Node.js, Express, PostgreSQL, Cloudflare Workers (Serverless Edge), REST/WebSocket APIs

Leadership & Methodologies: Technical Ownership, Architecture RFCs, Code Reviews, Mentoring, Agile/Scrum, Sprint Planning, European Market Localization (i18n), Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Senior Software Engineer — iOS | [Redpin \(Currencies Direct\)](#)

Mumbai, India • Aug 2023 – Oct 2025

Fintech — European International Money Transfer & FX iOS App

- **Architecture & Modularity:** Architected full-cycle iOS delivery from requirements through production using MVVM-C and Clean Architecture, reducing feature development cycle time by 25% through modular, reusable components.
- **Payment SDKs & Secure Transactions:** Integrated and optimized enterprise Payment SDKs and secure payment gateway flows (Apple Pay, card acquisition SDKs, 3D Secure / PSD2 Strong Customer Authentication), streamlining multi-currency transfers, ensuring PCI-DSS compliance, and eliminating payment drop-off across European corridors.
- **Performance Engineering:** Diagnosed execution bottlenecks and retain cycles using Xcode Instruments and telemetry, achieving a 35% reduction in production crashes and accelerating cold launch time by 30%.
- **App Store & User Growth:** Drove a 22% increase in monthly active users over 12 months, contributing to an App Store rating increase from 4.3 to 4.6 stars.

- **CI/CD & Release Automation:** Automated end-to-end testing, TestFlight validation, and App Store distribution pipelines via GitHub Actions, eliminating manual regression overhead.
- **European Localization (i18n):** Delivered end-to-end multi-language localization for 3 European languages, directly enabling regulatory compliance and market expansion across European financial corridors.
- **Technical Leadership & Mentorship:** Guided cross-functional architecture reviews, defined REST API contracts with backend teams, and mentored mid/junior engineers on SOLID principles, testability, and code quality.

Senior Software Engineer — iOS | [SUGAR Cosmetics](#)

Mumbai, India • Dec 2022 – Jul 2023

High-Growth Consumer E-Commerce iOS App (100K+ Active Users)

- **E-Commerce Feature Delivery:** Engineered high-conversion checkout funnels, product catalog browsing, and interactive bag flows, contributing to an 18% improvement in session duration.
- **Deep Memory Profiling & Optimization:** Investigated memory bloat using Xcode Allocations and Leaks Instruments; eliminated critical retain cycles, cutting memory footprint by 40% and binary size by 15%.
- **User Engagement & Deep Linking:** Integrated WebEngage push notification campaigns and Branch SDK deep linking, driving a 12% repeat-purchase uplift and 20% referral conversion growth.
- **Design System & Motion:** Translated Figma design systems into pixel-perfect UIKit/SwiftUI components and implemented lightweight Lottie animations, decreasing onboarding drop-off by 10%.

Software Developer — iOS | [PROVAB Technosoft \(Tap2Health Pro\)](#)

Bangalore, India (Remote) • Apr 2022 – Nov 2022

Telehealth & Doctor Consultation iOS Platform

- **Virtual Healthcare Consultations:** Built the doctor-facing iOS app supporting real-time virtual consultations via WebSockets, multi-mode video/voice/chat, and configurable schedule management.
- **Digital Signatures & e-Prescriptions:** Implemented secure e-prescription generation with Trustgate digital signature integration, automatically producing verified clinical PDF documents.
- **Doctor Verification & App Store Delivery:** Implemented multi-stage credential verification workflows and successfully deployed the Tap2Health Pro application to the Apple App Store.

Planned Career Break — Family Responsibilities

Nov 2019 – Mar 2022 • Dedicated Personal & Continuous Professional Upskilling Period

- Dedicated period for significant family responsibilities during COVID-19; maintained continuous technical currency in modern Swift, SwiftUI, Swift Concurrency (async/await), and iOS architecture before returning to full-time engineering.

Software Developer — iOS | [JAWK Softwares \(Neborhud\)](#)

Pune, India • May 2017 – Oct 2019

Residential Community iOS App (5,000+ Users)

- **Full-Lifecycle Mobile Development:** Built the Neborhud residential community iOS application from ground zero, supporting 5,000+ registered residents with events, polls, and real-time chat.
- **Automated Unit & UI Testing:** Created comprehensive XCTest test suites achieving 60% code coverage across core business logic, reducing regression defects by 30%.
- **Networking & App Store Operations:** Optimized networking layer with Alamofire request caching (slashing response latency by 20%), configured provisioning profiles, and managed App Store submissions.

Software Developer — PHP & Web | [CBIL360 \(InfiniteJourneys\)](#)

Pune, India • Sep 2016 – Jan 2017

- Engineered responsive web features, relational database operations, and REST API integrations for an online travel booking portal.

Associate Software Developer — Web & iOS | [The Next Interface Technology \(My Vote\)](#)

Kanpur, India • Sep 2014 – Jul 2016

- Developed native iOS applications using Objective-C and Cocoa Touch alongside web features, achieving a 95% defect-resolution rate across 3 client production releases.

KEY PRODUCTION PROJECTS & PRODUCT OWNERSHIP

Independent Engineering & Product Lab (Nov 2025 – Present) — Self-directed engineering initiative following Redpin tenure, architecting and shipping commercial-grade software across native Apple platforms (Swift 6, SwiftUI), real-time cross-platform multiplayer (iOS & Android), and SaaS.

MacMint — Native macOS Utility & Storage Optimizer | getmacmint.store

Tech: Swift 6, SwiftUI, AppKit, macOS 14+ (Sonoma), Cloudflare Workers, Dodo Payments
(Sep 2026 – Present)

- **Native macOS Architecture:** Architected and built a native macOS system utility in Swift 6 and SwiftUI, providing disk space analysis, deep developer cache purging (Xcode DerivedData, CocoaPods, Homebrew, Docker), and orphaned application uninstallation.
- **System Safety Guardrails:** Implemented strict safety guardrails (CleanerGuard) using immutable allow-lists and sandboxed file operations to prevent accidental deletion of critical system assets or user data.
- **Hierarchical Disk Visualizer:** Built an interactive radial Sunburst visualizer for real-time hierarchical storage breakdown across system drives and directories.
- **Serverless Licensing & Webhooks:** Integrated a serverless Cloudflare Workers edge microservice handling cryptographically verified webhook entitlements and email magic-link license activations via Dodo Payments.

TopBump — Pay-to-Rank SaaS & B2B Leaderboard Platform | topbump.lol

Tech: Next.js 16 (App Router), React 19, TypeScript, Prisma 7, PostgreSQL, Tailwind CSS v4, Vitest
(Aug 2026 – Sep 2026)

- **High-Concurrency Web Platform:** Engineered a high-concurrency ranking platform featuring a 65-category database-driven hierarchy, live bidding leaderboards, and rentable ad grid spaces.
- **Payment Abstraction Layer:** Architected a swappable PaymentProvider abstraction layer supporting pluggable payment processors alongside zero-dependency mock simulation for automated CI test suites.
- **B2B Admin & Domain Security:** Built an administrative portal for partner moderation, ad slot reservations, and domain safety verification, secured by cryptographically signed HTTP-only cookies.
- **Automated Transactional Testing:** Authored Vitest automated test suites ensuring atomic database transactions and strict ledger balancing under high concurrent updates.

DotMesh — Tactical Multiplayer Mobile Board Game | [App Store](https://apps.apple.com/us/app/dotmesh/id1644444444) • [Google Play](https://play.google.com/store/apps/details?id=com.dotmesh)

Tech: React Native, Expo, TypeScript, Firebase Realtime Database, App Store Connect, Google Play Console
(Apr 2026 – Jul 2026)

- **End-to-End Product Ownership:** Took independent end-to-end product and technical ownership from concept and UX through architecture, development, monetization, testing, store submission, and ongoing live operations.
- **Real-Time Multiplayer Networking:** Architected a real-time multiplayer networking layer with room code matchmaking and sub-second game-state synchronization using Firebase.
- **Adaptive Bot Heuristics:** Engineered offline vs. smart AI bot gameplay modes with scalable difficulty heuristics and custom grid-matrix state engines.
- **UI Aesthetics & Monetization:** Designed cyber-neon UI animations, responsive haptic feedback, and an integrated StoreKit/in-app subscription model for premium themes and ad-free play.
- **Dual App Store Release Management:** Fully managed production mobile releases across Apple App Store and Google Play, handling code signing, compliance, and App Store review requirements.

TinyOrbit URL — High-Performance SaaS URL Shortener & Analytics Portal | tiny-orbit-url.vercel.app

Tech: React 19, TypeScript, Vite, Node.js, Express, PostgreSQL, Recharts, JWT, Docker
(Jan 2026 – Mar 2026)

- **Decoupled API-First SPA:** Developed an API-first Single Page Application (SPA) decoupled from a Node.js/PostgreSQL RESTful service for lightning-fast branded link generation.
- **Real-Time Telemetry & Viz:** Built an interactive analytics dashboard with Recharts visualizing time-series click volumes, GeoIP geographic traffic distribution, and device attributions.
- **Enterprise Security & Hardening:** Enforced enterprise security protocols: JWT authentication, Google OAuth 2.0, HTTP-only cookies, Zod schema payload validation, and IP rate limiting to prevent API abuse.
- **REST Contract Architecture:** Designed scalable RESTful API contracts with standardized error handling, pagination, and OpenAPI-aligned schemas.

EDUCATION & CREDENTIALS

Bachelor of Technology (B.Tech) in Computer Science & Engineering — Allen House Institute, UPTU (71%) — 2013

Post Graduate Diploma in Advanced Computing (PG-DAC) — CDAC, IACSD Pune (71%) — 2014

Certifications: Preventing Financial Crime • Global Sanctions Compliance • Anti-Money Laundering • Anti-Bribery and Corruption (LRN) • Prompt Engineering (Udemy, 2025)